The empirical recurrence for OEIS A224646, proved by transfer matrix

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Abstract

OEIS A224646 counts the $(n+2)\times 5$ matrices over $\{0,\dots,1\}$ all of whose 3×3 contiguous subblocks are of equal population, and carries an empirical recurrence of order 20 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition ties together 3 consecutive rows and 3 consecutive columns at once, so a single row is not a state; the strip of the last 2 rows is, and the admissible strips are read off the overlap graph of the admissible windows instead of being enumerated. That makes $a(n)$ a walk count on $S = 2240$ vertices, hence C-finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation.

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1 The conjecture

OEIS A224646 is "Number of (n+2)X5 0..1 matrices with each 3X3 subblock having the same population.". It has offset 1 and begins

$$3200, 5696, 10304, 19328, 44096, 103232, 250304, 672320, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 4*a(n-1) -3*a(n-2) +36*a(n-3) -144*a(n-4) +108*a(n-5) -503*a(n-6) +2012*a(n-7) -1509*a(n-8) +3480*a(n-9) -13920*a(n-10) +10440*a(n-11) -12564*a(n-12) +50256*a(n-13) -37692*a(n-14) +22464*a(n-15) -89856*a(n-16) +67392*a(n-17) -15552*a(n-18) +62208*a(n-19) -46656*a(n-20)

As of the "Last modified" line on the live entry (Oct 03 2025 23:33:38, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

Call a 3×3 matrix over $\{0,\dots,1\}$ a *window* when it is of equal population, and call a 3×5 matrix a *band* when every one of its 3 contiguous 3×3 subblocks is a window. What the entry counts is the matrices of 5 columns whose every 3 consecutive rows form a band.

Take as states the 2×5 strips that occur as the top 2 rows of a band, together with those that occur as its bottom 2 rows, and put an edge from u to v when the band whose first 2 rows are u and whose last 2 rows are v exists. A band is determined by that pair, so the edge is unique when it exists. Reading a counted matrix from the top, its rows 1..2, then 2..3, and so on, form a walk, and every walk arises from exactly one matrix; a matrix with $n+2$ rows gives a walk of n edges. Since the first 2 rows of a counted matrix with at least 3 rows are the top of a band, and since no further condition falls on where a walk starts or ends,

$$a(n) \;=\; \iota^{\top} M^{\,n} \tau, \qquad \iota = \tau = (1, 1, \ldots, 1)^{\top},$$

with M the adjacency matrix on $S = 2240$ states. In particular a is C-finite. The alignment was checked against the entry's published terms rather than assumed.

There are $(1+1)^{10}$ strips of 2 rows and 5 columns, far too many to test one by one, and they are not tested one by one. Two consecutive windows of a band overlap in 3×2 entries, so a band is exactly a walk of 3 windows in the graph that joins a window to those whose first 2 columns are its last 2 columns. Enumerating those walks produces the bands, and with them the states and the edges, directly.

3 The admissible windows

The entry does not name the common value; it only asks that all the windows of one matrix agree. So fix a value v and let W_v be the set of 3×3 matrices over $\{0, \ldots, 1\}$ whose population, the number of its entries equal to 1, equals v. A matrix counted by the entry has all of its windows in a single W_v, and the value of v is determined by the matrix, so the sets of matrices belonging to different v are disjoint and

$$a(n) \;=\; \sum_{v} a_v(n),$$

each a_v being the walk count of Section 2 built from W_v alone. A disjoint union of the 10 digraphs, one per value of v, therefore counts the whole of $a(n)$ with the same all-ones vectors, and the 3×3 windows in play number 512 in total.

4 The proof

Lemma 1. *Let $q(t) = t^{20} - \sum_i c_i t^{20-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M) \tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+20} - \sum_i c_i M^{j+20-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) \;=\; 4\,a(n{-}1){-}3\,a(n{-}2){+}36\,a(n{-}3){-}144\,a(n{-}4){+}108\,a(n{-}5){-}503\,a(n{-}6){+}2012\,a(n{-}7){-}1509\,a(n{-}8){+}3$$

for every $n > 19$.

Proof. The vector $w = q(M)\tau$ was formed by 20 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 19 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A224646.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.